Skilling Up to Attempt the σ-Essential Problem

Foundations-first. A sequenced reading path with per-phase target-questions.

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Abstract

A sequenced, audit-structured study plan for the σ-essential contextual-state problem, written for the *foundations-first / path-agnostic* choice: build the common substrate—descriptive set theory, large cardinals, and the orthomodular-lattice σ-state corner—until the primary sources read fluently and the tractable edge reveals itself. Each phase carries **target-questions** tied to a specific section of the investigation, so the study has the same verify-against-a-question structure as the research. The driver is "learn and try"; the only real constraints are the theorems already proved (forcing-is-the-wrong-engine, masa-free-dead), not background.

1 The object, stripped to what a solver must touch

> A concrete σ-complete non-Boolean irreducible OML carrying a finite two-valued state on a $\perp$-closed subposet that extends to *no* global σ-additive two-valued state.

Localized: this is **σ-point-selection failure** (wall A)—a locally coherent two-valued pattern across overlapping incompatible blocks with no global σ-additive two-valued state. The frontier is **non-Polish / non-standard-Borel / irreducibly non-central**, the consistency strength is **genuinely unknown**, and **no borrowable principle exists in any surveyed field**. Resolution needs a genuinely new object—built, not transported.

The skill list is therefore *not* generic "learn set theory." It is targeted at the three things a witness-construction **or** an independence proof would demand, in dependency order.

The asymmetry (the reason for the ordering). The OML / quantum-logic cluster is already owned. The gaps are descriptive set theory and large cardinals. The plan front-loads those and treats the OML sources as practitioner-level re-reads.

Coverage / prerequisites (honest)

Every *field* below is covered by a **self-contained source**—no coverage hole. The caveats are *intra-book* ("read the early chapters, don't skip in"), not missing fields.

- **Kechris** is cold-startable from undergraduate real analysis + point-set topology + naive ordinals / cardinals; the plan jumps to Ch. 11, so if a Polish-space / tree construction feels unfamiliar, fall back to its *own* early chapters—not another book.
- **Jech** is the encyclopedic *reference*; its measurable-cardinal material leans on its *own* earlier chapters (ordinals, cardinals, cofinality, clubs / stationary sets, filters). Read those first—that is the real entry cost of Phase 2, not a separate text.

- **Operator algebra is deliberately scoped out.** Blecher–Weaver / Akemann–Weaver (masas, conditional expectations, pure states on von Neumann algebras) are a field not in any listed text—but the masa route is *proved dead*, and Phase 3 reads Blecher–Weaver only to *understand the boundary*, not build on it. So **light functional analysis ("what is a masa") suffices; deep operator-algebra study is not on the critical path.** Only a construction that needed to *enter* the masa route would reopen this—and the proved deadness says it won't.

Chapter numbers throughout are convenience pointers, not verified against current tables of contents—the **topic-based target-questions** are the robust part; navigate to the chapter by topic.

2 Phase 0 — the bridge you almost crossed (free, ∼1–2 weeks)

Before any new book, nail the **definability-vs-existence** distinction cold. It is the error *type* that killed the CBER bridge and it will recur at every DST contact.

- **Read:** the LB-mechanism refutation and the CBER kill with one lens— "no *Borel* selector" (a definability fact) vs. "no state *at all*" (an existence fact).
- **Target-question** (answer without notes): *why can a ZFC theorem (E_0, E_∞ non-smooth) neither force nor refute clause (ii)?* The answer— clause (ii)-failure is $\geq$ measurable strength; a ZFC fact is silent on measurable-strength statements; a definability obstruction cannot yield an existence obstruction—is the load-bearing intuition. The Lean polarity gate (`builds_state_implies_not_witness`) certifies this shape; read it too.

3 Phase 1 — Descriptive set theory (the witness's home regime, ∼2–3 months)

The frontier *is* the non-standard-Borel regime; Glimm–Effros is *your* dichotomy one abstraction level over; the whole DST sweep lives here.

Kechris, *Classical Descriptive Set Theory* (GTM 156)

Read by dependency, not linearly.

- **Ch. 11–14** — Borel / analytic / co-analytic hierarchy → **Borel equivalence relations** → the **Glimm–Effros dichotomy**.
  - *Target-Q:* state Glimm–Effros precisely ("a CBER is non-smooth $\iff$ E_0 embeds $\iff$ no Borel two-valued transversal"). Then: *which word in it does **not** transport to clause (ii)?* (Answer: "Borel.") This closes the CBER lead with understanding, not on say-so.
- **Ch. 16–18** — uniformization, the projective hierarchy.
  - *Target-Q:* co-analytic uniformization-failure was DST-corner #1. *Why is its obstruction real-valued / over a standard-Borel base—i.e. why is it Derr–Williamson-killed?*

Gao, *Invariant Descriptive Set Theory*

Read **after** Kechris supplies the language. The E_0 / E_∞ / hyperfinite / turbulence book in one place.

- *Target-Q:* define hyperfinite; state Adams–Kechris (uncountably many non-Borel-bireducible CBERs). Then the one that matters: *why is E_∞ being non-hyperfinite* ***orthogonal*** *to clause (ii)?* (Non-hyperfiniteness buys more non-smoothness; non-smoothness is exactly what is orthogonal to an existence obstruction.) When this is obvious, the DST corner is genuinely closed for you.

4 Phase 2 — Large cardinals and forcing (the strength axis, ∼2–3 months)

The bare object on the Boolean shadow *is* a measurable cardinal; the open strength question is whether non-distributivity pushes above or below it.

Jech, *Set Theory* (3rd millennium ed.)

- **Ch. 8** — filters, ideals, ultrafilters. The substrate for everything below.
  - *Target-Q:* why is $\mathcal{F}_s = \{A : s(A) = 1\}$ **not** a filter on a non-Boolean concrete OML, even for a Dirac state? (The MO_2 unit test: $a \cap b \neq \varnothing$ as sets yet $a \wedge b = 0$ lattice-wise.) This is your carrier's defect stated in filter language.
- **Ch. 10** — measurable cardinals; **Ulam matrices**.
  - *Target-Q:* re-derive the Ulam matrix argument by hand. Locate the exact line where the construction needs **non-orthogonal unions / comprehension**— that line is why the matrix argument does not transplant to an orthogonal-join-only OML.
- **Ch. 14–15** — forcing, specifically to *feel* **Lévy–Solovay**.
  - *Target-Q:* why does set-forcing preserve measurability (small forcing can neither create nor destroy it)? This is *why forcing is the wrong engine* for the existence direction—you need the mechanism, not the citation.

Kanamori, *The Higher Infinite* (second pass, only if the strength path calls)

Ch. 1–2 (measurable → ultrapower → elementary embedding $j : V \to M$, critical point κ). Only after Jech. *Target-Q:* the UB route would "transport U along j"—but the measurable gives U on **Boolean** $\mathcal{P}(\kappa)$; transport to orthogonal-only is the disjointification wall from the construction side. Understand why that is a costume.

5 Phase 3 — your own primary sources, read as a practitioner (∼1–2 months, interleaved)

By now you read these for **technique**, not verdict.

- **Navara–Pták 1983** — re-derive their ZFC non-Dirac σ-state construction. This is your **existence-proof template** and the object that refuted the LB mechanism. *Target-Q:* what makes their carrier $L_{f,g}$ support a non-Dirac global σ-state—and why is it **not** a witness (the two compatible observables / "Boolean enough" point)?

- **Derr–Williamson 2023 + Maharam 1972 §8** — the Polish upper boundary. *Target-Q:* exactly where is inner-regularity load-bearing in Thm D.6 / Maharam §8.2? (This is the boundary a witness must escape.)
- **Blecher–Weaver 2017** — the Hilbert analogue that does **not** transfer. *Target-Q:* identify the ultrafilter-extraction-from-a-masa step (Prop. 2.3); state the **masa-free question** it leaves open for the concrete two-valued case.

6 Where to start today

Kechris Ch. 12–14 (Borel equivalence relations → Glimm–Effros), paired with the Phase-0 re-read. It is the single chapter that makes the frontier-map legible **and** is the most reusable across all three resolution paths. After ∼a month you will feel whether to lean DST-construction or strength-axis—the path-agnostic plan working as intended.

7 Honest constraints (the only real ones)

- **Theorems, not background.** Forcing-wrong-engine (Lévy–Solovay) and masa-free-dead are *proved* dead ends—respect those. Everything else is learn-and-try; there is no ceiling.
- **The convergence is the finding so far.** This plan is for *attempting* the open object, not re-litigating the closed literature frontier. If a phase surfaces a borrowable principle the sweep missed, that is a real result—but expect to *build*, not find.

Companion: `oml_lead_philosophical_reading.md` (the question's phenomenological framing); `sigma_essential_taxonomy.json` (zoom-out attempt index); `sigma_essential_large_cardinal_bounds.md` (the bounds and every section-reference above). Markdown source: `sigma_essential_skill_plan.md`.